

Questions

1	Define the term programming paradigm. (1 mark)
2	Explain the difference between object-oriented programming and procedural programming. (2 marks)
3	Explain the difference between a class and an object. (2 marks)
4	Identify a situation where a static method may be used. (1 mark)
5	Using pseudocode, write a class with relevant attributes and methods to represent a digital clock object. It should show the time, and update the time at the end of each minute. (6 marks)

1. Programming Paradigm – Different methods of planning how to complete a task through programming
2. Difference between object oriented and procedural programming – OOP focuses on real life application, focusing on specific 'objects' that have connections between each other whereas procedural programming is declaring variables that are not related to each other
3. A Class contains objects, acting as their 'blueprint' whereas an object is an individual member of a class
4. If a pins in a class need to share information between each other
5. :

Class clock

```
class Clock
```

```
    private hours
```

```
    private minutes
```

```
    private seconds
```

```
    public procedure set(newHour,newMinutes)
```

```
        this.hours = newHour
```

```
        this.minutes = newMinutes
```

```
    endprocedure
```

```
public procedure display(hours,minutes)
    output(hours,minutes)
end procedure
```

```
private procedure time(hours,minutes,seconds)
    if seconds = 60 then
        set(hours, minutes +1)
    endif
endprocedure
```

```
public procedure new(hours,minutes)
    display(hours,minutes)
end procedure
```

```
endclass
```